

Mathematics (GOAT)

Basic Mathematics

Lecture-II

Concepts Covered in the Class:

- Properties RECAP
- Characteristic & Mantissa of Log
- Graphs of Logarithm
- Graphs of Exponential Functions

Properties RECAP

- Logarithms simplify complex calculations through standard algebraic rules, such as turning multiplication into addition ($\log_b(xy) = \log_b x + \log_b y$) and exponents into coefficients ($\log_b(x)^y = y \log_b x$).
- Base conversion and fundamental exponent identities are essential tools to remember, for example: $b^{\log_b N} = N$ and $\log_b N = (\log_c N)/(\log_c b)$.
- If the base itself has an exponent, it factors out as a reciprocal: $\log_{b^k} N = (1/k) \log_b N$.



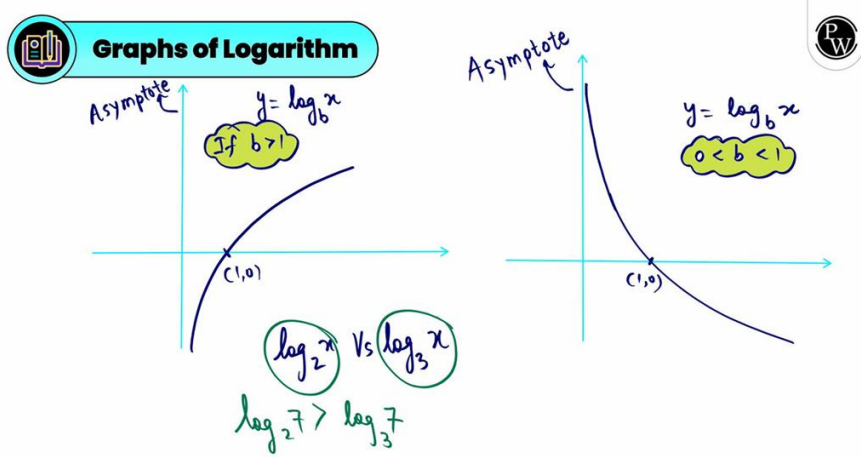
Characteristic & Mantissa of Log

- Any logarithmic value can be separated into an integer part (the Characteristic, I or Ch) and a positive decimal part (the Mantissa, f).
- $\log x = I + f$ (where $0 \leq f < 1$)
- When dealing with negative logarithm values, you must adjust the integer downwards so the mantissa remains positive (e.g., $-3.71 = -4 + 0.29$). Furthermore, for base 10 logarithms, the total number of digits in a number N is exactly one more than the characteristic of $\log_{10} N$.

Graphs of Logarithm

- The fundamental graph of $y = \log_b x$ always crosses the x -axis at the point $(1, 0)$ and approaches the y -axis as a vertical asymptote.
- The direction of the curve depends strictly on its base: it is a continuously increasing curve if $b > 1$, and a continuously decreasing curve if $0 < b < 1$.
- The function is strictly “one-one,” with a domain limited only to positive real numbers (R^+) and a range that spans all real numbers (R).





Graphs of Exponential Functions

- The exponential curve $y = a^x$ is the inverse concept, always crossing the y-axis at $(0, 1)$ and using the x-axis as its horizontal asymptote.
- Just like logarithms, the curve represents exponential growth (increasing) when the base $a > 1$ (such as e^x , where $e \approx 2.71$), and exponential decay (decreasing) when $0 < a < 1$.

